

DATA/CS 172: Python for Data Analysis

Understanding Types

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Announcements

- Lab 1 due today
- Reminder for lab 2: Careful not to hold files open too long!
 - Also, probably not good to open/close files more than we need to
 - Opening/closing a file takes time

Agenda

- Variable Types
- String basics
- Type conversion

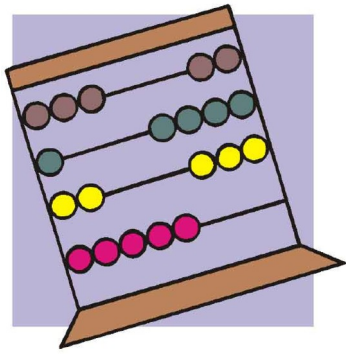
Variable Types

Variable Types

- So far, our variables have mostly represented numbers...
 - Remember from math that there are different types of numbers
 - Integers, real Numbers, Complex Numbers, Rational Numbers etc.
- In python, there are three common basic **types** for a variable ←
Important term!
 - int, float, str
 - There are others (booleans, complex types), we'll talk about later



int



- Short for integer
- Examples:
 - -1
 - 367
 - 3000
- Note: Must omit commas! 3,000 not an integer!
- Basic arithmetic (+, -, *, /) work as expected
 - Until you start working with really large numbers

float

- [illegible]

A large, stylized black pi symbol (π) with a subtle drop shadow, centered on a light blue background.

3.14159

265358979323

84626433832795

0288419716939937.

105820974944592307
16406006000000000000

16406286208998628
348253421170679821480

65132823066470938446095505822

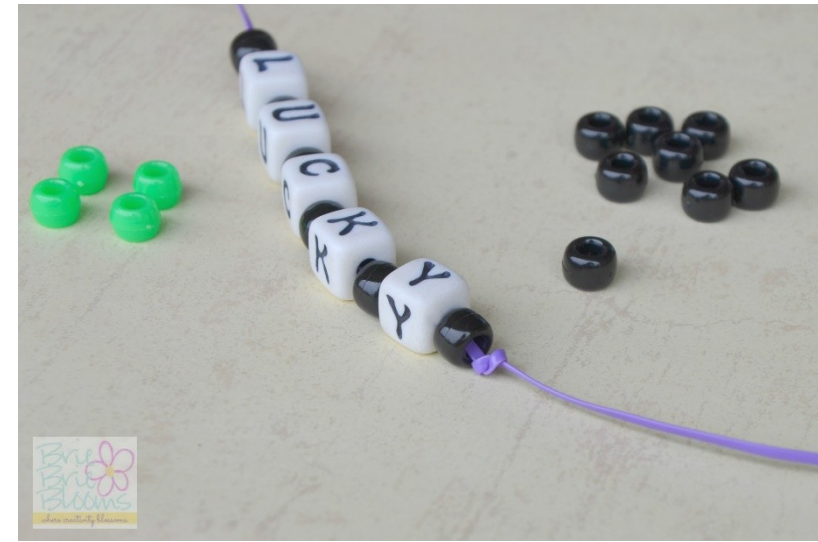
596446229489549303819644288109756659334461284756482337
7831652712019091456485669234603486104543266482133936072602491412737

[illegible]

42522384045346680552619318817010003137838752886855320283810264871776691304530825340428755482358785
5778187780532171220866612001927876611195909216240129938095257214056786628786593615338182796823031502350318529089757362259041
97217523479135155748572425451506959582853311068172535880750983877463749439192550040092770161139090948824012858361635767060140711018194295559
76783784255353774726847104047304610280468425906949129337677028091521047521605696024508038101935152333835840427406473269149127260426992279678235785

str

- Short for “String of Characters”
 - Example –
“hello” is a string of 5 characters
- Those things enclosed in quotes
 - Unlike some languages, fine to enclose in single(') or double(“) quotes
 - If you have quotes inside your string, you need to be careful here
 - ‘Here’s a string’ ← Not okay
 - “Here’s a string” ← Okay
 - ‘This is a “nice” string’ ← Okay
 - “Here’s a “nice” string” ← Uh oh, how do I fix this?!



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Strings and special characters

- Backslash allows you to escape special characters like quotes
- Also allows you to put other characters in string
 - Line break (aka newline) - `\n`
 - Tab - `\t`
 - Can also be used with quotes
 - Demo
- Hmm, haven't we shifted the problem back? What if I want to put a backslash in my string?
 - Right - `\\`
- Much more you can do with strings – we'll talk about this later
 - If you work with text data all this will be key



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How does Python know which type my variable is?

- Based on the value!
 - What type is 3?
 - Right! So if I write `a = 3`, Python knows the type of `a` is `int`!
 - How about `a=3.14`?
 - How about `a="hello world"`?
 - How about `a="3.14"`?
 - How about `a=hello`?



Note: unlike other languages...

- Python's type system is fairly weak
 - Almost “anything goes”
- A variable's type is not fixed!
- `a=2`
- `a="hello"`
 - Perfectly fine!
 - In a strongly typed language like C++, this would never work!
 - Usually bad style, but switching between floats and ints is fine
 - Good rule of thumb: If in doubt, **add more variables**
 - Keeps code readable



How to convert types?

- In python, **type conversion** is very easy ← Important term!
- `nameOfType(variableYouWantTochange)`
- Example:
 `a = "1"`
 `b = int(a) + 2`
 `print(b) //prints 3`



Why convert?

- Operators mean different things depending on type!
- Great example: +
 - Demo
 - + on numbers is called addition (duh)
 - + on strings is called string **concatenation** ← Important term!
 - Different strings joined together
 - Careful: comma , seems similar but only works in print()
 - Demo
- Another reason: data!
 - By default, values are read from files as strings
 - May need to convert them to numbers
 - By default, can only write strings to files
 - Need to use the str() cast to convert whatever we have to a string

Things can go wrong when converting...

- `int("hello")`
- Demo
- We get an error: how can we handle this?
- **Error handling** with `try/except`! ← Important term!

`try:`

```
    int("hello")
```

`except ValueError:`

```
    print("error converting to int!")
```

- Often we will want to exit on errors
- `import sys` at the top
- `sys.exit(exitcode)` (demo)
 - Usually an **exit code** of 0 means success, anything else failure ← Important term!



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Lab 3 out

- Feel free to use lab 2 as a starting point